University of Florida, Department of Mathematics

Algebra First-Year Exam, August 2025, Part 2

Answer 4 questions. Write your solutions clearly in complete English sentences. You may quote theorems (within reason) as long as you state them clearly.

1. Let R be a commutative ring with $1 \neq 0$.

(a) Define what it means for an ideal of R to be maximal.

(b) Prove that every proper ideal of R is contained in a maximal ideal.

2. This problem has two parts.

(a) Define the term Euclidean domain.

(b) Prove that the polynomial ring $k[x]$, where k is a field, is a Euclidean domain.

3. This problem has two parts.

(a) Give the definition of the term unique factorization domain.

(b) Show that the ring $\mathbb{Z}[\sqrt{-5}]$ is not a unique factorization domain.

4. Let k be a field, let $k[x,y]$ denote the polynomial ring in two variables over k. Let $R = k[x,y]/(xy)$ and denote the image in R of a polynomial $f \in k[x,y]$ by $\bar{f}$.

(a) Prove that if P is a prime ideal of R then either $\bar{x} \in P$ or $\bar{y} \in P$ (or both).

(b) Describe all of the prime ideals of R that contain $\bar{y}$.

5. Using the rational canonical form, find one representative for each conjugacy class in $\mathrm{GL}(5, \mathbb{F}_2)$ whose elements have order 4.

6. Let L be an algebraic field extension of the field K and let R be a subring of L that contains K. Prove that R is a field.